

Zhili WANG

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EDUCATION

The Hong Kong University of Science and Technology

Ph.D. in Data Science and Analytics; GPA: 3.74/4.0; Supervisor: Lei CHEN

Hong Kong SAR, China

Sep. 2020 – Feb. 2025

The Hong Kong University of Science and Technology

M.S. in Big Data and Technology; GPA: 3.70/4.0

Hong Kong SAR, China

Sep. 2019 – Jun. 2020

University of Electronic Science and Technology of China

B.Eng. in Spatial Information and Digital Technology; GPA: 3.87/4.0

Chengdu, Sichuan, China

Sep. 2015 – Jun. 2019

RESEARCH INTERESTS

- **Quantitative Research:** Investigating the full lifecycle of systematic trading strategies.
- **Graph Neural Networks and Relational Modeling:** developing novel architectures for relational data modeling with applications in both financial industry and academic research.
- **Autonomous AI and Neural Architecture Search:** exploring effective automated neural network design and architecture optimization under computational constraints.

ACADEMIC PUBLICATIONS

- **Zhili Wang**, Shimin Di, Lei Chen, Xiaofang Zhou. *Search to Fine-tune Pre-trained Graph Neural Networks for Graph-level Tasks* (ICDE 2024).
- **Zhili Wang**, Shimin Di, Lei Chen. *A Message Passing Neural Network Space for Better Capturing Data-dependent Receptive Fields* (SIGKDD 2023).
- **Zhili Wang**, Shimin Di, Lei Chen. *AutoGEL: An Automated Graph Neural Network with Explicit Link Information* (NeurIPS 2021).
- Hanmo Liu, Shimin Di, Jialiang Wang, **Zhili Wang**, Jiachuan Wang, Xiaofang Zhou, Lei Chen. *Structuring Benchmark into Knowledge Graphs to Assist Large Language Models in Retrieving and Designing Models* (ICLR 2025).
- Jialiang Wang, Shimin Di, Hanmo Liu, **Zhili Wang**, Jiachuan Wang, Lei Chen, Xiaofang Zhou. *Computation-friendly Graph Neural Network Design by Accumulating Knowledge on Large Language Models* (arXiv Preprint 2024).

PROFESSIONAL EXPERIENCE

Quantitative Researcher Intern

Y-Intercept (Hong Kong) Limited

Hong Kong SAR, China

Jan. 2025 – Jul. 2025

- **Network Momentum Strategy for Multi-Asset Futures**

- Developed similarity-based sparse relational matrices across EMEA futures to aggregate features via graph message passing, capturing alpha from momentum spillovers.
- Boosted Sharpe from 0.8 to 1.3 and reduced MDD_r from 6% to 3% over OLS (OOS: 2020-present).

- **Transformer-Based Intraday FX Strategy**

- Engineered feature groups from minute-level data across G10 and EM FX pairs during peak hours.
- Developed Transformer model to extract cross-asset lead-lag alphas with volatility-adjusted sizing.
- Achieved Sharpe > 3.5 with moderate signal correlation (0.2-0.4) between LASSO and Transformer models (accumulated rolling OOS: 2009-2020).

- **Tree-Based Quarterly Earnings Prediction for US Equities**

- Engineered fundamental features to forecast quarterly changes in key metrics (net income, EPS).
- Implemented tree models with error-based position sizing to optimize risk-adjusted returns.
- Achieved Sharpe > 1.3 and MDD_r < 8% (OOS: 2022-present).

SKILLS SUMMARY

- **Programmings:** Python, PyTorch
- **Languages:** Native Mandarin, Proficient English (IELTS 7.5, GRE 319+3)
- **Personal Interests:** Strength Training, Culinary Arts